

Transformers, power flow analysis part 2

ELEC0447 - Analysis of electric power and energy systems

Bertrand Cornélusse

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Overview I

1. The Transformer

2. Transformers in the power flow analysis

What will we learn today?

- ▶ The power transformer: physical models, nameplate test data ($v_k\%$, $v_{kr}\%$), and per-unit representation
- ▶ Delta-wye connections, phase shifts, and standard vector groups (IEC clock convention)
- ▶ Tap changers (OLTC) and phase-shifting transformers (PST) for voltage and power flow control
- ▶ How to include transformers in power flow analysis:
 - ▶ Two-port admittance matrix and system $\mathbf{Y}_{\text{bus}}$ stamping
 - ▶ Equivalent π -model for in-phase regulation ($t \in \mathbb{R}$)
 - ▶ Loss of $\mathbf{Y}_{\text{bus}}$ symmetry for phase shifters ($t \in \mathbb{C}$)
- ▶ Hands-on 4-bus case study with PST in PandaPower and Newton-Raphson

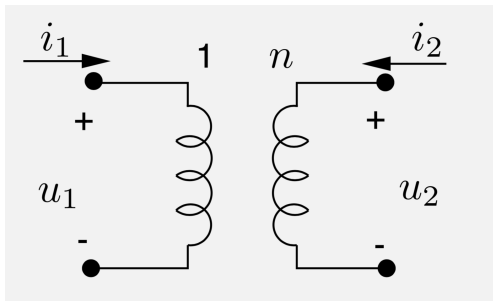
You will be able to do exercises 6.2, 6.3, 6.4, 6.18, 6.19 from Ned Mohan's book.

The Transformer

Basic model of a transformer

A (single phase) transformer is made of two magnetically coupled **coils** or **windings**. An ideal transformer is a two-port represented as

with



$$u_2 = nu_1$$

$$i_2 = -\frac{1}{n}i_1$$

Sign convention note: Currents i_1 and i_2 both enter the positive terminals (standard two-port convention), explaining the minus sign. With load convention (i_2 leaving), $i_2 = \frac{1}{n}i_1$.

Usages

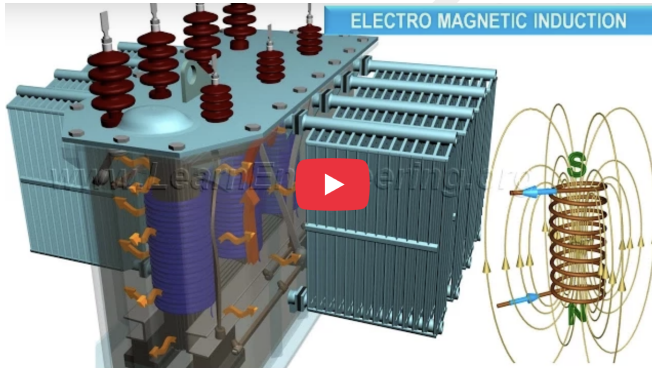
In power systems, transformers are mainly used to transmit power over long distances by changing the voltage level, thus decreasing the current for a given power level. The voltage level of a synchronous generator is around 20kV.

Voltage is changed around five times between generation and load.

It is also used to

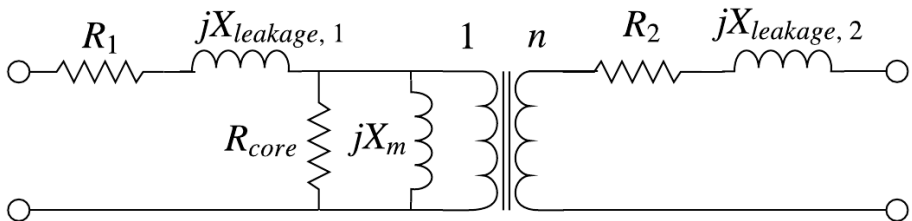
- ▶ **measure** currents and voltages,
- ▶ electrically **isolate** parts of a circuit (not the auto-transformer we will see),
- ▶ and **match** impedances.

How does a transformer work?



[Link to the video](#)

Non-ideal model I



The ideal model is complemented by elements

Non-ideal model II

- ▶ X_m that models the magnetizing inductance
- ▶ $X_{leakage,i}$ that models the flux not captured by the core on side i
- ▶ R_{core} that models eddy current and hysteresis losses, i.e., losses in the iron core
- ▶ R_1 and R_2 that model (coil) copper losses

Parameters are either given in the datasheet or obtained by open-circuit and short-circuit tests.

The core is usually **laminated** to decrease losses.

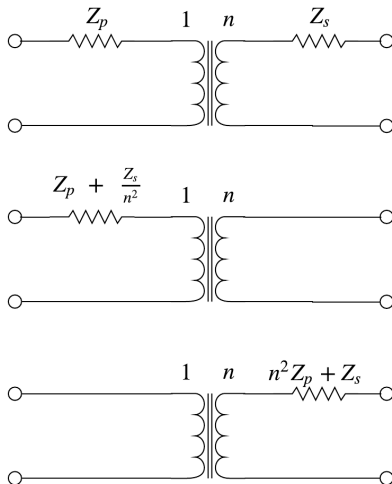
Simplification

The excitation current, the sum of the currents in R_{core} and X_m , is often neglected, leading to a simpler non-ideal model, and the series impedances can be transferred from one side to the other, with

$$Z_p = R_1 + jX_{leakage,1}$$

and

$$Z_s = R_2 + jX_{leakage,2}$$



Per unit representation I

Let's consider the rated voltages and currents on both sides of the (ideal) transformer as base values. As

$$V_{s,base} = nV_{p,base}$$

and

$$I_{p,base} = nI_{s,base},$$

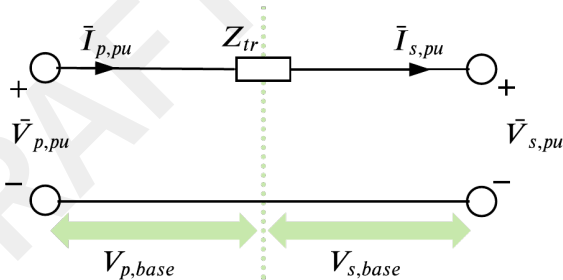
the **MVA base is the same on both sides**, and thus

$$Z_{s,base} = n^2 Z_{p,base}$$

Per unit representation II

Hence, **in per unit, the transformer can be replaced by a single impedance**

$$Z_{tr} = \frac{Z_p}{Z_{p,base}} + \frac{Z_s}{Z_{s,base}}.$$



Per unit representation III

Thus we have also that

$$\begin{aligned} Z_{tr} &= \frac{Z_p + Z_s/n^2}{Z_{p,base}} \\ &= \frac{n^2 Z_p + Z_s}{Z_{s,base}} \end{aligned}$$

i.e. **the impedance is the same whether we see it from the primary or the secondary side, although the voltage bases differ.**

Also, if the three-phase transformer is **wye-delta** connected, a **30° phase shift** must be applied (more on this later).

From Nameplate Data to Per-Unit Parameters I

In manufacturer datasheets and power flow software (e.g., PandaPower), transformers are specified via standard factory test values:

- ▶ **Ratings:** rated power S_n [MVA] and rated voltages V_{n1}, V_{n2} [kV]
- ▶ **Short-circuit test (copper losses & leakage):**
 - ▶ Short-circuit voltage: $v_k\%$ (typically 8% to 18% for transmission units)
 - ▶ Ohmic drop / copper losses: $v_{kr}\%$ (or short-circuit active power P_{sc} in kW)
- ▶ **Open-circuit / no-load test (iron losses & magnetizing):**
 - ▶ No-load active losses: P_{fe} [kW] (or $p_0\%$)
 - ▶ No-load magnetizing current: $i_0\%$ (typically $< 0.5\%$, usually neglected in power flow)

From Nameplate Data to Per-Unit Parameters II

1. On the transformer's own rated base (V_n, S_n):

$$z_{k,\text{tr}} = \frac{v_k\%}{100} \text{ pu}, \quad r_{k,\text{tr}} = \frac{v_{kr}\%}{100} = \frac{P_{\text{sc}}}{S_n} \text{ pu}, \quad x_{k,\text{tr}} = \sqrt{z_{k,\text{tr}}^2 - r_{k,\text{tr}}^2} \text{ pu}$$

2. Scaling to the system network base ($V_{\text{base}}, S_{\text{base}}$):

$$Z_{k,\text{sys}} = (r_{k,\text{tr}} + jx_{k,\text{tr}}) \times \left(\frac{S_{\text{base}}}{S_n} \right) \times \left(\frac{V_n}{V_{\text{base}}} \right)^2$$

Since $X_k \gg R_k$ in large power transformers (often $X/R > 20$), R_k is often neglected in simplified transmission calculations ($Z_{\text{tr}} \approx jx_k$).

From Nameplate Data to Per-Unit Parameters III

PandaPower Implementation

In PandaPower, this conversion is computed directly from parameters:

$$z_k = (vk_percent/100) \times (net.sn_mva/sn_mva)$$

$$r_k = (vkr_percent/100) \times (net.sn_mva/sn_mva)$$

$$x_k = \sqrt{z_k^2 - r_k^2}$$

General Per-Unit Base Conversion Formula

When converting an impedance from $(V_{\text{base}}^{\text{old}}, S_{\text{base}}^{\text{old}})$ to a new base $(V_{\text{base}}^{\text{new}}, S_{\text{base}}^{\text{new}})$:

$$Z_{\text{pu}}^{\text{new}} = Z_{\text{pu}}^{\text{old}} \times \left(\frac{S_{\text{base}}^{\text{new}}}{S_{\text{base}}^{\text{old}}} \right) \times \left(\frac{V_{\text{base}}^{\text{old}}}{V_{\text{base}}^{\text{new}}} \right)^2$$

- ▶ When voltage bases across transformer zones match the nominal turns ratio:

$$\frac{V_{\text{base},1}}{V_{\text{base},2}} = \frac{V_{n1}}{V_{n2}} \implies \frac{V_{\text{base}}^{\text{old}}}{V_{\text{base}}^{\text{new}}} = 1$$

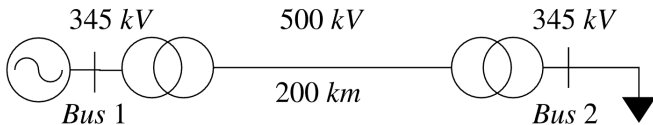
- ▶ Then Z_{pu} scales **solely with the MVA base ratio**:

$$Z_{\text{pu}}^{\text{new}} = Z_{\text{pu}}^{\text{old}} \times \left(\frac{S_{\text{base}}^{\text{new}}}{S_{\text{base}}^{\text{old}}} \right)$$

This property is what makes per-unit analysis so powerful across multiple voltage tiers.

Illustration of Per-Unit normalization I

This is Example 6.1 from the reference book. Consider the one-line diagram



with

- ▶ a 200 km line with $R = 0.029\Omega/km$, $X = 0.326\Omega/km$, neglected shunt impedances
- ▶ two transformers with a leakage reactance of $0.2pu$ in the (500 kV, 1000 MVA) base, and losses neglected.

Illustration of Per-Unit normalization II

What is the equivalent model in a (345 kV, 100 MVA) base?

In the (500 kV, 1000 MVA) base:

- ▶ $Z_{line,pu} = 200 \times (0.029 + j0.326)/(500^2/1000) = 0.0232 + j0.2608pu$
- ▶ hence, the total impedance between buses 1 and 2 is

$$Z_{12} = 0.0232 + j0.2608 + 2 * j0.2pu = 0.0232 + j0.6608pu$$

In the (345 kV, 100 MVA) base

- ▶ the pu value of the impedance is the same in the (500 kV, 1000 MVA) and (345 kV, 1000 MVA) bases!

Illustration of Per-Unit normalization III

- ▶ Why? since we can transfer the impedance from one side of each transformer to the other, cf. a previous remark.
- ▶ if we now change the MVA base to 100 MVA,

$$Z_{12} = (0.0232 + j0.6608) \times (100/1000)pu = 0.00232 + j0.06608pu$$

since the base impedance is proportional to the inverse of the MVA base.

Efficiency

The efficiency expressed in % is

$$100 \times \frac{P_{output}}{P_{input}} = 100 \times \left(1 - \frac{P_{losses}}{P_{input}} \right)$$

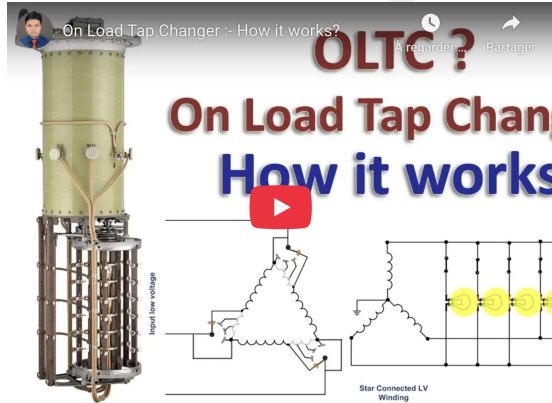
Remarks:

- ▶ maximal when loaded such that copper losses = iron losses (cancel derivative of efficiency w.r.t current)
- ▶ Around **99.5 %** in large power transformers at full load.

Tap changers

- ▶ Some transformers are equipped with a system allowing the $1 : n$ ratio to be changed
- ▶ The ability to change the tap under load is called load tap changer (LTC) or on-load tap changer (OLTC)
- ▶ This is mainly used for voltage control
- ▶ It is usually implemented using auto-transformers
- ▶ We will see later on how to include this in the power flow analysis

Tap changing transformers



[Link to the video](#)

Auto-transformers: Working Principle

The two windings are connected in series on the same core, sharing a common section without galvanic isolation.

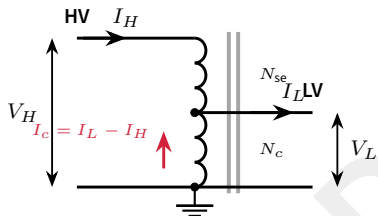
Voltage & Current Relations:

$$\frac{V_H}{V_L} = 1 + \frac{N_{se}}{N_c}, \quad I_c = I_L - I_H$$

Dual Power Transfer:

- ▶ **Inductive** (magnetic): $S_{ind} = S_{rated} \left(1 - \frac{V_L}{V_H}\right)$
- ▶ **Conductive** (direct): $S_{cond} = S_{rated} \left(\frac{V_L}{V_H}\right)$

When $V_H/V_L \approx 1$, the core is sized only for $S_{ind} \ll S_{rated}$, making it much smaller and lighter!



Auto-transformers: Trade-offs & Applications

Advantages:

- ▶ Physically smaller and lighter
- ▶ Less costly (significantly less copper & iron)
- ▶ Higher efficiency ($\eta > 99.7\%$)
- ▶ Lower leakage reactance
- ▶ Easy to implement tap changers (OLTC)
- ▶ Common solid earth ground

Disadvantages:

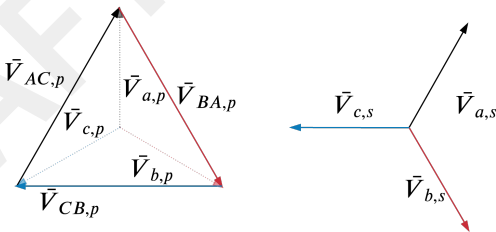
- ▶ No galvanic / electrical isolation
- ▶ Higher short-circuit current
- ▶ Risk of full V_H on secondary if common winding opens
- ▶ Economic benefit diminishes if ratio $V_H/V_L > 2$

Typical Applications: Interconnecting close transmission voltage levels (e.g. 380/220 kV, 220/150 kV) and serving as the regulating unit in on-load tap changers and phase shifters.

Phase shift in delta-star transformers

The star part has n times the number of turns of the delta part (primary side).
Let's reason on phase a ,

- ▶ Voltage $\bar{V}_{a,s}$ is on the same core as $\bar{V}_{AC,p} = \sqrt{3}\bar{V}_{a,p}\angle -30^\circ$ where $\bar{V}_{a,p}$ is the (virtual) phase-neutral voltage on the primary side.
- ▶ Since $\bar{V}_{a,s} = n\bar{V}_{AC,p}$,
 $\bar{V}_{a,s} = n\sqrt{3}\bar{V}_{a,p}\angle -30^\circ$



We **gain a $\sqrt{3}$ factor** in the amplification, and a **lagging phase shift of 30°** .

The same reasoning holds for phases b and c .

Standard Three-Phase Vector Groups (IEC 60076)

In industrial practice, phase displacement between windings is designated by a **clock number** (each hour represents a 30° lag):

Notation Rules:

- ▶ 1st letter: HV winding (**D** = Delta, **Y** = Wye)
- ▶ 2nd letter: LV winding (**d** = delta, **y** = wye, **n** = neutral out)
- ▶ Number $k \in \{0, \dots, 11\}$: phase lag of LV line-to-neutral phasor relative to HV:
$$\theta_{LV} - \theta_{HV} = -k \times 30^\circ$$

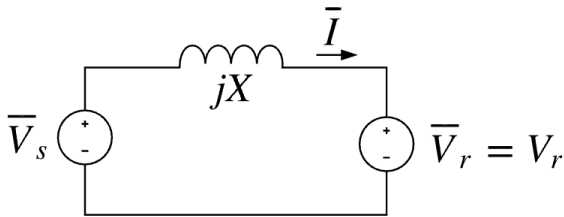
Common Examples:

- ▶ **Yy0 / Dd0**: 0° shift (12 o'clock)
- ▶ **Dy1**: Delta-Wye, LV lags by 30° (1 o'clock)
— *the case studied on previous slide!*
- ▶ **Dy11** (or **Yd11**): LV leads by 30° / lags by 330° (11 o'clock)
- ▶ **Yd5**: LV lags by 150° (5 o'clock)

Paralleling Condition: Transformers can be operated in parallel only if they belong to the **same vector group** (identical phase displacement).

Power flow regulation by phase shifting I

We have seen that active power flows are dictated by the voltage magnitudes but also the sine of the angle difference between buses:



$$S_r = \bar{V}_r \bar{I}^* = V_r \left(\frac{V_s \angle -\delta - V_r}{-jX} \right)$$

$$= \frac{V_s V_r \sin \delta}{X} + j \frac{V_s V_r \cos \delta - V_r^2}{X}$$

δ is the angle between $\bar{V}_r$ and $\bar{V}_s$

Power flow regulation by phase shifting II

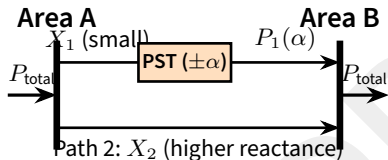
If we have a device that can generate an adjustable phase shift, we can control the power flows. This is the purpose of **phase-shifting transformers**.

In practice phase shifting is achieved by "combining the signal with a fraction of itself shifted by 90° ". For the details of how it is implemented or modeled, see

- ▶ Wikipedia
- ▶ Section 5.7 of Weedy or ELEC0014.
- ▶ ENTSO-E - Phase Shift Transformers Modelling, Version 1.0.0, May 2015

Physical Principle: Controlling Parallel Path & Loop Flows

Active power divides inversely proportional to line reactances: $P \sim \Delta\theta/X$. In meshed grids, this creates **transit flows** and line overloads.



Without PST ($\alpha = 0$):

$$P_1 = P_{total} \frac{X_2}{X_1 + X_2} \quad (\text{overload on Line 1!})$$

With PST ($\alpha \neq 0$):

$$P_1 \approx \frac{\theta_A - \theta_B + \alpha}{X_1}$$

Tuning α redirects power into Path 2, relieving Line 1 without costly generation redispatch!

Example: phase shifting transformers on the borders of Belgium I



380/380 kV, in series with:

1. line Zandvliet (B) - Borssele (NL)
and Zandvliet (B) - Geertruidenberg (NL)
2. line Meerhout (B) - Maasbracht (NL)
3. line Gramme (B) - Maasbracht (NL)
 - ▶ nominal power
 $3V_N I_{\max} = 1400 \text{ MVA}$
 - ▶ phase shift adjustment: 35
positions, $+17 / -17 \times 1.5^\circ$ (at
no load)

Example: phase shifting transformers on the borders of Belgium II

220/150 kV :

- ▶ in series with the Chooz (F) - Monceau (B) line nominal power: 400 MVA
- ▶ in-phase adjustment: 21 positions, $+10/ - 10 \times 1.5\%$
- ▶ quadrature adjustment: 21 positions, $+10/ - 10 \times 1.2^\circ$

Remarks

In **three-phase operation**,

- ▶ either there are three separate single-phase transformers (easier to fix when there is a problem on a phase, more modular)
- ▶ or a **three-phase transformer**, that is a single core with three auto-transformers on it, cf. the video at the beginning of this presentation (cheaper, lighter core and less copper).

Some transformers called **three-winding transformers** are equipped with a third winding (not to be confused with a three-phase transformer) that is used for auxiliary purposes (feeding auxiliary devices e.g., fans, providing reactive power support, ...).

Transformers in the power flow analysis

Transformer without regulation

A transformer, in the per-unit representation, can thus be represented

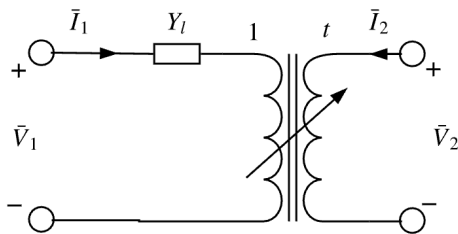
- ▶ as a two-port if the shunt admittance is considered
- ▶ as a simple series leakage impedance if the shunt admittance is neglected

It is a bit harder to model controllable taps for voltage and flow control.

Representing taps and phase shifts I

Let Y_l be the leakage admittance and t be the off-nominal turns ratio:

- ▶ If $t \in \mathbb{R}$ (typically $0.90 \leq t \leq 1.10$, step size $\approx 0.5\%-1.25\%$), this corresponds to an **in-phase tap-changer (OLTC)** for voltage magnitude and reactive power control.
- ▶ If $t = me^{j\theta}$ with $\theta \neq 0$ ($|\theta| \leq 15^\circ-25^\circ$), this is a **phase-shifting transformer (PST)** for active power flow control.



Representing taps and phase shifts II

We have

$$\bar{I}_1 = \left(\bar{V}_1 - \frac{\bar{V}_2}{t} \right) Y_l$$

and since $\frac{\bar{V}_2}{t} \bar{I}_1^* = -\bar{V}_2 \bar{I}_2^*$ by energy conservation

$$\bar{I}_2 = -\frac{\bar{I}_1}{t^*} = -\bar{V}_1 \frac{Y_l}{t^*} + \bar{V}_2 \frac{Y_l}{|t|^2}$$

Thus tap and phase shift can be represented by the two-port admittance matrix:

$$\begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} Y_l & -\frac{Y_l}{t} \\ -\frac{Y_l}{t^*} & \frac{Y_l}{|t|^2} \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix}$$

System $\mathbf{Y}_{\text{bus}}$ Stamping & Newton-Raphson Impact I

For a transformer connected between **Bus i** (series impedance side) and **Bus j** (ideal tap side 1 : t):

$$\begin{bmatrix} \bar{I}_i \\ \bar{I}_j \end{bmatrix} = \begin{bmatrix} Y_l & -\frac{Y_l}{t} \\ -\frac{Y_l}{t^*} & \frac{Y_l}{|t|^2} \end{bmatrix} \begin{bmatrix} \bar{V}_i \\ \bar{V}_j \end{bmatrix}$$

System $\mathbf{Y}_{bus}$ Stamping & Newton-Raphson Impact II

Stamping Rule into the Global $\mathbf{Y}_{bus}$ Matrix:

$$Y_{ii} \leftarrow Y_{ii} + Y_l$$

$$Y_{jj} \leftarrow Y_{jj} + \frac{Y_l}{|t|^2}$$

$$Y_{ij} \leftarrow Y_{ij} - \frac{Y_l}{t}$$

$$Y_{ji} \leftarrow Y_{ji} - \frac{Y_l}{t^*}$$

System $\mathbf{Y}_{\text{bus}}$ Stamping & Newton-Raphson Impact III

Consequence for Complex Taps ($t = me^{j\theta}, \theta \neq 0$)

Because $t \neq t^*$, the off-diagonal terms differ:

$$Y_{ij} \neq Y_{ji} \implies \mathbf{Y}_{\text{bus}} \neq \mathbf{Y}_{\text{bus}}^T \quad (\text{loss of symmetry!})$$

Direct Impact on the Newton-Raphson Power Flow:

- Standard textbook formulations often assume $Y_{km} = Y_{mk}$
($G_{km} = G_{mk}, B_{km} = B_{mk}$).

System $\mathbf{Y}_{\text{bus}}$ Stamping & Newton-Raphson Impact IV

- ▶ With PSTs, one **must** strictly use directed branch admittance values:

$$P_k = \sum_{m=1}^n V_k V_m (G_{km} \cos(\theta_k - \theta_m) + B_{km} \sin(\theta_k - \theta_m))$$

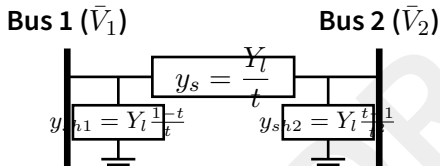
where $G_{km} + jB_{km} = Y_{km} \neq Y_{mk}$.

- ▶ Off-diagonal blocks in the Jacobian $\mathbf{J}$ are no longer transpose-symmetric.

The Equivalent π -Model for In-Phase Taps ($t \in \mathbb{R}$)

When $t \in \mathbb{R}$ (pure voltage magnitude regulation, no phase shift), $t = t^*$, so $Y_{ij} = Y_{ji}$. The two-port is reciprocal and can be synthesized as a **passive π -equivalent**:

Equivalence with Matrix:



$$Y_{11} = y_s + y_{sh1} = Y_l$$

$$Y_{22} = y_s + y_{sh2} = \frac{Y_l}{t^2}$$

$$Y_{12} = Y_{21} = -y_s = -\frac{Y_l}{t}$$

Advantage: Standard transmission line π -routines can be directly reused!

Note on PST ($t \in \mathbb{C}$): Non-reciprocal ($Y_{12} \neq Y_{21}$), cannot be modeled by passive two-terminal elements.

Illustration: The 4-Bus System with PST I

In the course exercises, we place a PST between Bus 1 and an auxiliary Bus 4 connected to Bus 3:

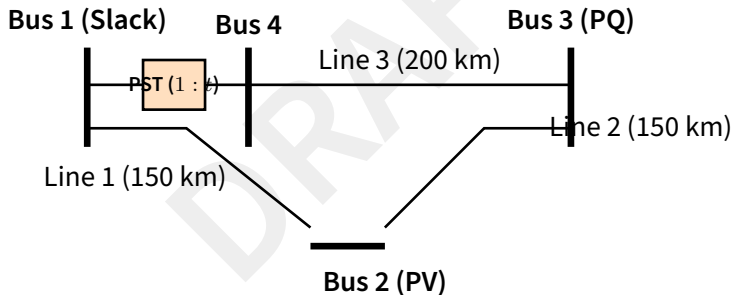


Illustration: The 4-Bus System with PST II

Effect of PST Tap Shift (from -8° to $+8^\circ$):

Tap Angle	Line 1	Line 2	Line 3
-8°	61.2 %	43.1 %	98.5 %
0°	75.4 %	68.2 %	84.1 %
$+8^\circ$	92.8 %	89.4 %	67.3 %

- ▶ As tap angle increases, active power is diverted from Line 3 towards Path 1-2.
- ▶ Line 3 is relieved without changing generator outputs!

Illustration: The 4-Bus System with PST III

Network Reduction (Kron / Norton Equivalent): Since no power is injected at auxiliary Bus 4 ($I_4 = 0$):

$$\mathbf{Y}_{\text{reduced}} = \mathbf{Y}_{aa} - \mathbf{Y}_{ab} \mathbf{Y}_{44}^{-1} \mathbf{Y}_{ba}$$

This yields an exact 3×3 equivalent network for custom Newton-Raphson solvers.

Python Notebooks & Implementation

► PandaPower Notebook

► Newton-Raphson Script

Summary: Branch Modeling in Power Flow

Element	Ratio / Parameter	$\mathbf{Y}_{\text{bus}}$ Symmetry	Passive π -Model	Main Control
AC Line	$Z = R + jX$	Symmetric	Yes	Passive
Fixed Transfo	1 : 1 (in pu)	Symmetric	Yes	Voltage level
OLTC	1 : $t, t \in \mathbb{R}$	Symmetric	Yes	Voltage / Q
PST	1 : $me^{j\theta}, \theta \neq 0$	Non-symmetric	No	Active flow P

Key Takeaways

- ▶ Standard line routines can represent OLTCs via modified π -shunts.
- ▶ Phase shifters require explicit directed admittance stamping in $\mathbf{Y}_{\text{bus}}$ and Newton-Raphson Jacobians.